

How Much of "Real Estate" Growth Is Just Pretend Rent?

Data sources: OECD National Accounts (SNA/ESA) and OECD Analytical House Price Indicators. **Coverage:** Up to 59 countries, spanning 1990–2024 where available. **Methods:** SQL-based data engineering (DuckDB), trend regression, paired hypothesis testing, correlation analysis, gradient-boosted machine learning (XGBoost), explainability analysis (SHAP), a benchmark neural network, and walk-forward-validated forecasting

Abstract

Every year, government statisticians report how much the "real estate sector" contributed to the size of the economy. But a large and rarely-discussed part of that number is not made of rent checks, agent commissions, or property-management fees; it is **imputed rent**: a notional "rent" that a homeowner is scored as paying to themselves, simply for living in a house they already own. No cash changes hands, no tenant exists, and no job is created by it, yet it is booked as economic output.

This report asks a simple question with a not-so-simple answer: how big is this "pretend rent" component, is it growing, what drives it, and can it be predicted? Using national accounts data for up to 59 countries (with a reliable, cross-checked panel of 43), we find that imputed rent typically makes up about two-thirds (median ~65%) of everything counted as "real estate sector output" — and that this is not a new or fringe phenomenon, but a permanent, structural feature of how real estate is measured almost everywhere.

We also find that whether this share is rising or falling is highly country-specific — it is rising in Czechia and the Netherlands, falling in the UK and Australia, and essentially flat in the US — so there is no single global story. However, looking only at *recent* growth (since 2019), imputed rent has driven the *majority* of nominal real-estate-sector growth in every flagship country studied (58.5% in the US, 73.8% in Czechia, 85.5% in the UK) — meaning that much of the "real estate boom" reported since the pandemic is, numerically, a valuation effect rather than new market activity.

Finally, we show that year-to-year swings in this "imputed share" cannot be reliably predicted from housing-market conditions using linear regression, gradient boosting, or a neural network — all three approaches fail to beat the

simple guess of "no change," implying that big jumps are usually caused by one-off statistical reclassifications rather than market cycles.

The report is written for a general audience: renters, homeowners, real-estate professionals, policymakers, journalists, and everyday readers of economic news — not just economists.

1. Introduction

1.1 Why this matters to you

Picture two headlines published on the same day:

"Real estate contributed \$50 billion to GDP growth this year"

"The average person pays \$0 more this year for the actual services agents, landlords, property managers, and developers provide"

Both of these headlines can be true at the same time — and this report explains why, and what that means for you, whether you rent, own, invest in, or sell property.

The reason is a piece of accounting most people have never heard of: **imputed rent**. When you own the home you live in, you don't pay yourself rent. But government statisticians still want to measure the economic value of "living in a house" consistently, whether you own it or rent it — otherwise a country where everyone suddenly bought their rented home would appear to get *poorer* overnight, purely on paper, even though nothing real changed. Their fix: estimate what you *would* pay if you rented your own home on the open market, and count that imaginary payment as part of the economy's output.

This is defensible accounting. But it also means that a large chunk of what gets reported under the banner "real estate sector" every year is not made of actual transactions — no tenant, no agent, no job, no invoice. It is a valuation exercise performed by statisticians, not a market event.

1.2 The research question

This report asks, and answers with data:

1. **How big** is the imputed-rent share of real-estate output, in a typical country?
2. **Is it rising or falling** — and is that the same everywhere, or does it vary by country?
3. What is actually **driving recent real-estate "growth"** headlines — new market activity, or rising notional valuations?
4. Does the imputed share **track real-world indicators** like house prices and actual rents?
5. **Can next year's move be predicted** from today's housing-market conditions?
6. What can we **honestly forecast** about where this is heading by 2030?

1.3 A translation table for the terms used in this report

Every section below uses precise technical language — that precision is what makes the numbers trustworthy — but each technical term is explained here in plain, everyday words, and every major finding is followed by an **"In everyday terms"** box that restates it without jargon.

Technical word	What it actually means, in plain language
Imputed rent	The "pretend rent" a homeowner is scored as paying to <i>themselves</i> . No cash changes hands, but statisticians still write down a rent payment for it and count it as economic output.
Real estate activities (excl. imputed)	The <i>real</i> , money-changing-hands side of the property business: agents' commissions, letting fees, property management, developers building and selling.

Technical word	What it actually means, in plain language
Value added / GDP	The overall size of the economic pie for a year — everything of value the country produced, added up.
Share / percentage point (pp)	If something makes up "60%" of a total, that is its <i>share</i> . If the share moves from 60% to 65%, that is a rise of 5 percentage points , not "5%" (which would mean a much smaller move, 60→63).
CAGR (compound annual growth rate)	The steady yearly growth rate that would carry a number from its starting value to its ending value — like the interest rate on a savings account that compounds every year.
Nominal vs. real ("volume")	<i>Nominal</i> = the raw currency figure, including the effect of rising prices. <i>Real</i> = with rising prices stripped out, showing whether the actual <i>amount of activity</i> grew.
Median	Line every country up from lowest to highest — the median is the one standing exactly in the middle. It is a more honest "typical value" than an average, which can be dragged around by one extreme country.
Correlation (r)	A score from -1 to +1 for "do these two things move together?" +1 = perfectly together, 0 = unrelated, -1 = perfectly opposite. $r = 0.30$ means "somewhat related, but plenty else is going on."
Statistical significance (p-value)	Roughly, "the odds this pattern is just random noise." $p < 0.05$ is the usual bar for "probably a real pattern, not a fluke."
SHAP value	A way of asking a prediction model "which ingredient gets credit or blame for this specific answer?" — like a chef explaining which spice made a dish taste a certain way.
Forecast confidence band	The shaded zone around a forecast line — an honest way of saying "our best guess is the middle line, but anywhere in this shaded range wouldn't surprise us."

2. Data and Methodology

2.1 What data was used

The analysis draws on the OECD's official National Accounts database, which — for many but not all countries — separately reports two components of the "Real estate activities" industry (ISIC classification code L68):

- **L68A — Imputed rents of owner-occupied dwellings:** the notional, non-market value described above.
- **L68B — Real estate activities excluding imputed rents:** brokerage, letting, property management, and real-estate development services — genuine, market-priced activity.

This was cross-checked against the OECD's Analytical House Price Indicators (house price and rent price indices) to see whether the accounting numbers track what is actually happening in housing markets.

2.2 Coverage

- The raw dataset covers **1,044 country-year rows across 59 countries**, with individual country series ranging from a single reported figure up to 35 years of history (Czechia: 1990–2024; the United States: 1997–2024; the United Kingdom: 1997–2023).
- After removing years where a component was not separately reported (see Section 4 for why this matters), a **reliable panel of 519 country-year rows across 43 countries** remains, forming the basis for every share and trend calculation in this report.

2.3 Analytical approach

Stage	Method
Data cleaning & structuring	SQL-based data engineering (DuckDB)
Statistical inference	Trend regressions, paired hypothesis tests (t-test, Wilcoxon), correlation analysis

Stage	Method
Machine learning	XGBoost vs. linear regression vs. a "no-change" naive baseline, tested with grouped cross-validation (so no country is tested on data it was trained on)
Explainability	SHAP (SHapley Additive exPlanations), to see which factors a model leans on
Deep learning	A compact neural network, honestly benchmarked against the simpler methods above
Forecasting	Walk-forward-backtested trend extrapolation to 2030, with each country's method chosen by which approach actually performed best historically

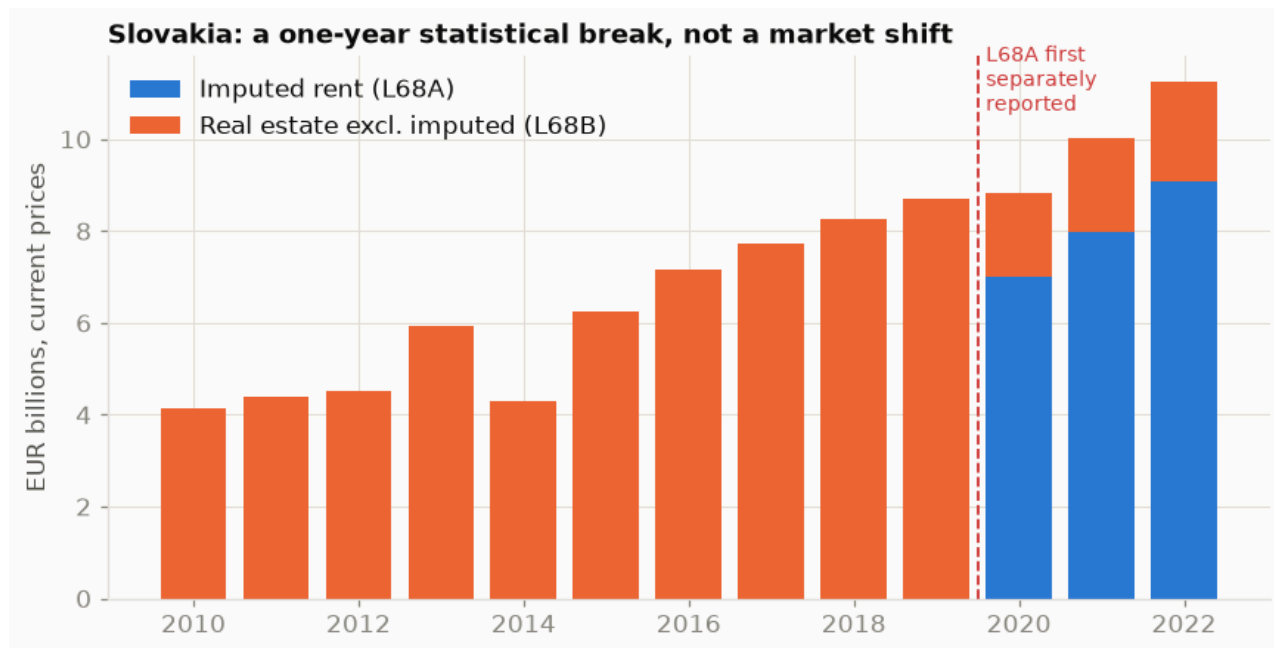
All results below are computed directly from the underlying data — nothing is estimated by hand.

3. Result 1 — A Cautionary Tale: Not Every "Zero" Is Real

Before trusting any trend in this data, we had to solve a data-quality trap: a country reporting "0" for imputed rent almost never means the actual value was zero — homeowners exist in every country, so imputed rent is never truly zero. It almost always means the statistical office simply hadn't started separately reporting it yet.

Slovakia illustrates this perfectly.

Figure 1. Slovakia: a one-year statistical break, not a market shift



For 2010–2019, Slovakia reported real estate activity as one blended number (shown in orange). Then, in 2020, its statistical office began publishing the imputed-rent component separately —and it turned out that imputed rent instantly made up about 79–81% of the total. This wasn't a housing boom. It was a bookkeeping change: statisticians started counting something that was always happening (people living in homes they own) but had never been measured separately before.

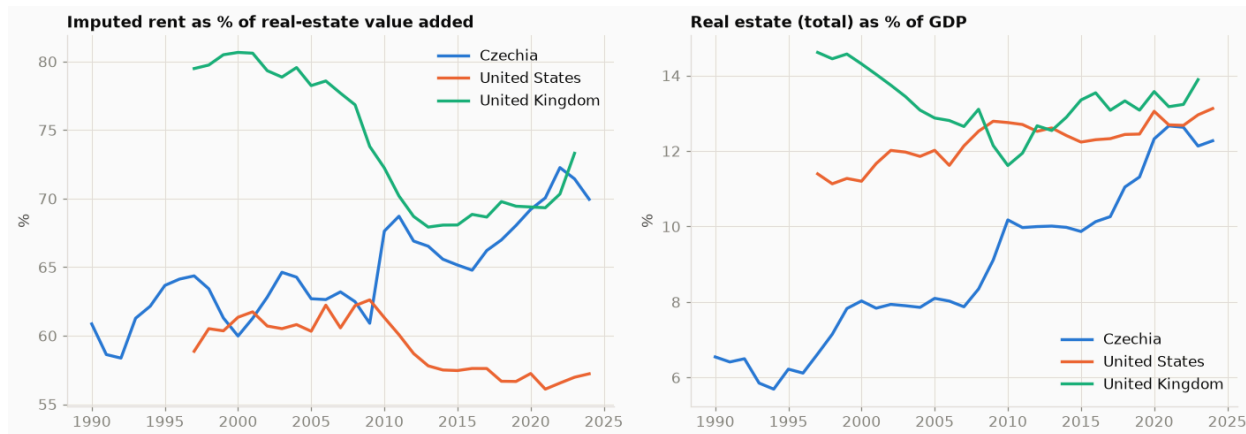
Imagine your child's school never graded "handwriting" before — then one term, out of nowhere, handwriting counts for 80% of their overall grade. Their report card suddenly looks transformed. Did their handwriting actually get 80% better overnight? No — the school just *started counting something it always ignored*. Slovakia's real-estate numbers did exactly this. Anyone glancing at the total that year would think Slovakia had a massive property boom. It didn't — the ruler changed, not what was being measured.

This is why the analysis below only uses years where both components were positively and separately reported — a "reliable panel" of 519 country-year observations across 43 countries.

4. Result 2 — The Three Flagship Countries: Czechia, the US, and the UK

Czechia, the United States, and the United Kingdom have the longest, most continuous data (35, 28, and 27 years respectively), long enough to see genuine long-run trends rather than noise.

Figure 2. Imputed rent share and real estate's overall size, 1990–2024



Country	Period	Imputed share at start	Imputed share at end	Change (percentage points)
Czechia	1990–2024	60.9%	70.0%	+9.1 pp
United Kingdom	1997–2023	79.5%	73.3%	–6.2 pp
United States	1997–2024	58.9%	57.2%	–1.6 pp

This means Czechia's "pretend rent" share of its real-estate sector has been *rising* over three decades — from about 61 cents to 70 cents of every dollar counted. The United Kingdom moved in the *opposite* direction: its imputed share *fell*, because its genuine market services (letting agencies, brokers, property managers) grew even faster than the imputed component. The US sits roughly flat. There is no single global story here — the direction of travel is country-specific.

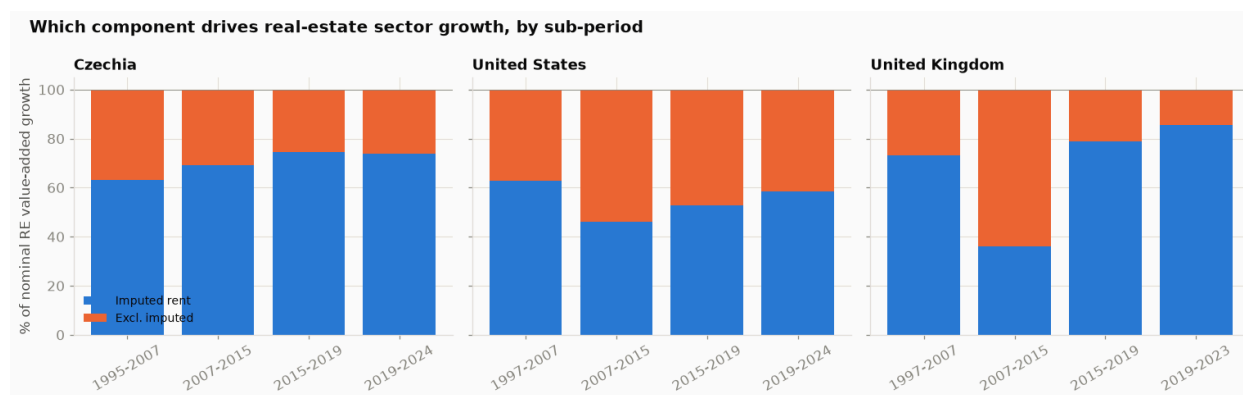
Picture two market towns. In Czech Town, out of every \$100 the "property business" is credited with, about \$61 used to be pretend homeowner rent — now it's \$70. The real, money-in-hand side (agents, letting,

building) grew too, just more slowly. In London Town, it's the reverse: the pretend-rent slice actually *shrank* from about \$80 to \$73 out of every \$100, because agents, landlords, and developers picked up the pace faster than pretend rent did. So if a neighbor in Czechia and a neighbor in the UK both tell you "our real estate sector grew," they may be describing two *different kinds* of growth.

5. Result 3 — What's Actually Driving *Recent* Growth Headlines?

Long-run trends can hide what's happening *right now*. So we asked a sharper question: of the real estate sector's growth in each recent period, how much came from rising "pretend rent" values versus real market activity?

Figure 3. Which component drives real-estate sector growth, by sub-period



Country	Period	Total sector growth	Share of that growth from imputed rent	Share from genuine services
Czechia	2019–2024	+49.7%	73.8%	26.2%
United States	2019–2024	+43.8%	58.5%	41.5%
United Kingdom	2019–2023	+31.6%	85.5%	14.5%

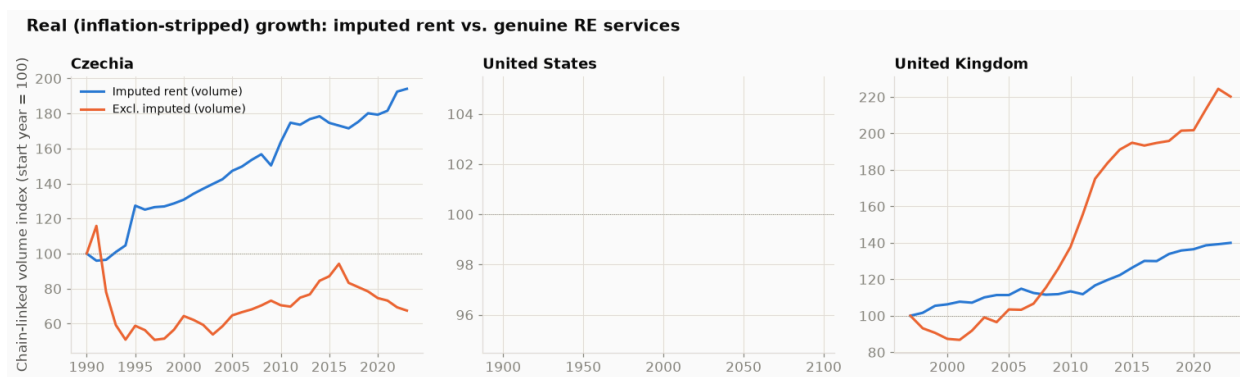
This means that since 2019, a majority of every country's reported real-estate-sector growth came from rising imputed-rent valuations, not new market transactions. In the UK, an extraordinary 85.5% of recent "growth" is

imputed rent; in Czechia, 73.8%; even in the US, where the imputed share of the *level* is roughly flat over the long run, 58.5% of recent growth is still imputed. This is true even in the UK, where the long-run trend of the imputed share is falling — meaning the recent post-pandemic period behaves differently from the long-run average. If your neighbor says "the property sector grew a lot these past few years," a fair follow-up question is: *how much of that was actually agents doing deals and builders building, versus the accountants' pretend-rent number simply climbing?* Based on this data, in most countries right now, well over half — often 70–85% — of that "growth" is the second kind.

6. Result 4 — Stripping Out Inflation: Who's *Really* Growing?

Nominal growth (raw currency figures) can be misleading because prices rise every year anyway. Stripping out inflation ("real" or "volume" growth) shows whether actual *activity* — not just its price tag — has grown.

Figure 4. Real (inflation-adjusted) growth: imputed rent vs. genuine real-estate services



- **Czechia (1990–2023):** imputed rent grew **+2.03% per year in real terms**, while genuine, non-imputed real-estate services actually **shrank by –1.19% per year**.
- **United Kingdom (1997–2023):** the picture flips — genuine services grew *faster* (**+3.08%/yr**) than imputed rent (**+1.30%/yr**).

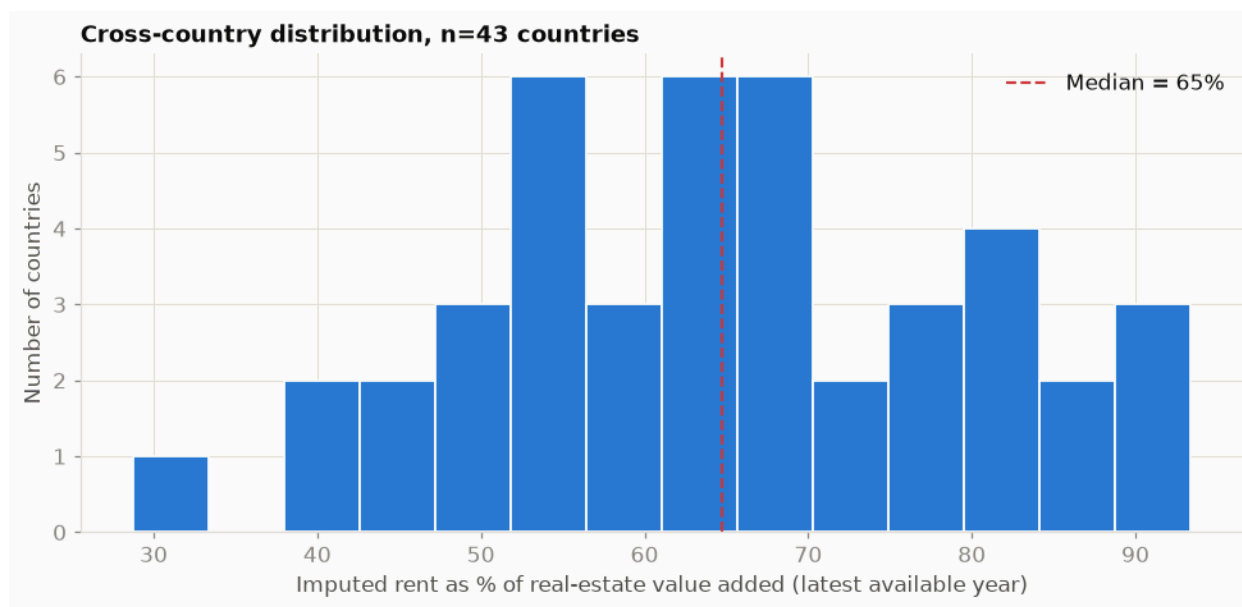
This is arguably the single most important chart in this report. In Czechia, once you strip away rising prices, the country's *real, working* property-services industry — brokers, letting agents, developers — has actually been **contracting**. All the headline "growth" is a valuation effect layered on top of a shrinking genuine sector. In the UK, the opposite is true: real market activity is outpacing the valuation effect, meaning UK real-estate GDP figures may, if anything, be *understating* genuine market dynamism.

Think of a market stall selling both fresh vegetables (real sales, every day) and a "display shelf value" the stallholder writes down for produce they'd theoretically sell to themselves. In Czechia, actual vegetable sales — real brokers, real property managers, real construction — have quietly been shrinking once you account for rising prices; all the reported "growth" is the display-shelf number climbing. In the UK, it's the opposite: real trading is outpacing the display-shelf number. This is the one chart that tells you which kind of "growth" you're really looking at.

7. Result 5 — Is This a Global Trend, or Country-by-Country?

Widening the lens to all 43 countries with reliable data:

Figure 5. How large is the imputed share across 43 countries?

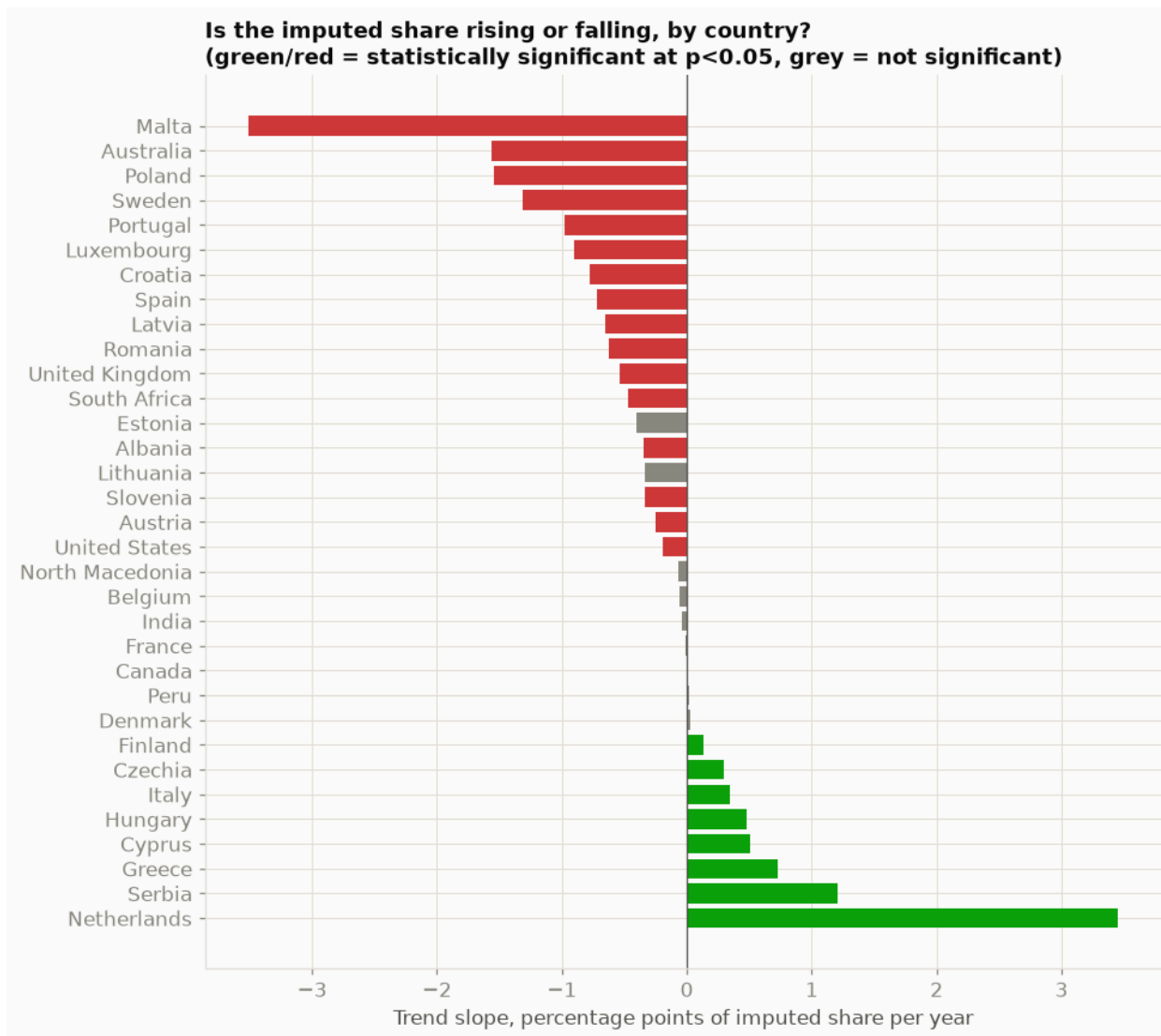


- **Median across 43 countries: ~65%**
- **Middle half of countries: roughly 54%–77%**
- Range: from about 29% to about 93%

In a *typical* country, about two-thirds of everything counted as "real estate sector output" is a rent nobody actually paid. This is not a recent drift or a quirk of a few countries — it is a permanent, structural feature of how real estate is measured almost everywhere.

If you lined up 43 countries from "least pretend-rent-heavy" to "most," the country standing exactly in the middle counts pretend rent as roughly 65 cents of every dollar in its real-estate numbers. That's not one unusual country — it's the typical case. So the next time a news report says "the property sector added \$10 billion to the economy this year," a reasonable first guess is that around \$6.50 of every \$10 of that was never actually paid by anyone to anyone.

Figure 6. Is the imputed share rising or falling, by country?



- **8 of 33** long-series countries show a statistically significant **rising** imputed share (Netherlands showing by far the steepest rise, at roughly +3.4 percentage points per year; followed by Serbia, Greece, Cyprus, Hungary, Italy, Czechia, and Finland).
- A similar number show a significant **falling** trend, led by Malta (falling roughly -3.5 percentage points per year), followed by Australia, Poland, Sweden, Portugal, Luxembourg, Croatia, Spain, Latvia, Romania, the United Kingdom, South Africa, Albania, Slovenia, Austria, and the United States.
- The rest (Estonia, Lithuania, North Macedonia, Belgium, India, France, Canada, Peru, Denmark) show no statistically reliable trend at all — essentially flat.

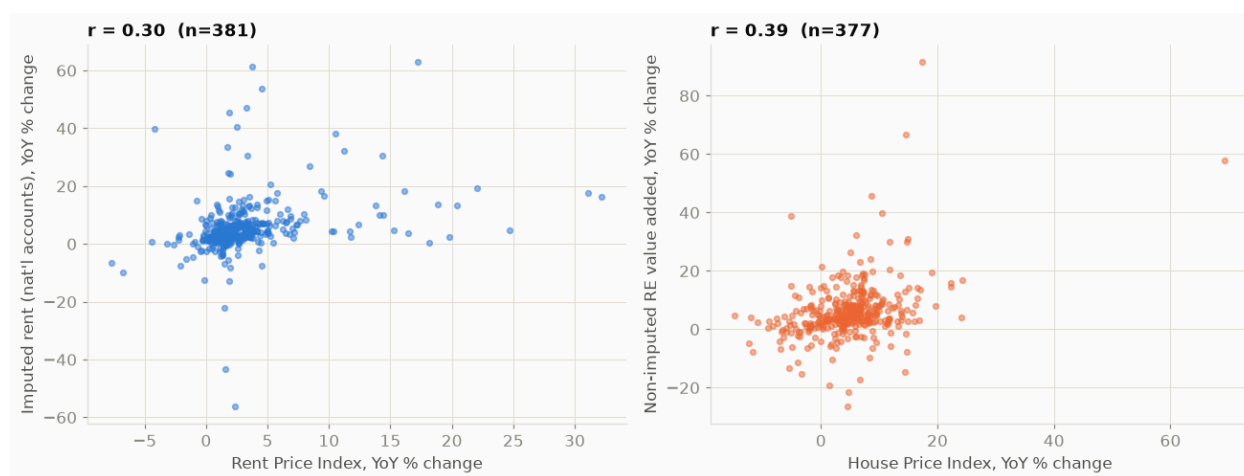
A formal statistical test across all 32 countries with long-enough histories (comparing each country's own early years against its own later years) found no significant average movement in either direction (t-test $p \approx 0.23$; Wilcoxon test $p \approx 0.057$, borderline). In plain terms: the "rising pretend-rent" story is real for some countries, but it is not a universal global trend — it cancels out on average against countries moving the opposite way.

In everyday terms, two extreme countries dominate this picture: tiny Malta, where the pretend-rent share has been rapidly *shrinking* (probably reflecting a genuine boom in real rental/tourist property activity), and the Netherlands, where it has been rapidly *growing*. Most other countries move much more modestly in one direction or the other, and a meaningful group show no clear direction at all. Don't believe any claim that "housing GDP is becoming fake everywhere" — the honest answer is "it depends which country you're in."

8. Result 6 — Does the "Pretend Rent" Number Track Reality?

If imputed rent purely tracked real market rents, its year-to-year growth should closely match the OECD's independently-compiled Rent Price Index (real rents that real tenants pay). Similarly, genuine (non-imputed) real-estate activity should track the House Price Index, since agents' and brokers' fee income scales with transaction values.

Figure 7. Imputed rent vs. rent prices, and genuine real estate vs. house prices



Comparison	Sample size	Correlation (r)	Statistically significant?
Imputed rent growth vs. official Rent Price Index growth	381	0.30	Yes
Non-imputed real estate growth vs. House Price Index growth	377	0.39	Yes
Imputed share level vs. house-price-to-income ratio	400	-0.06	No (p = 0.21)

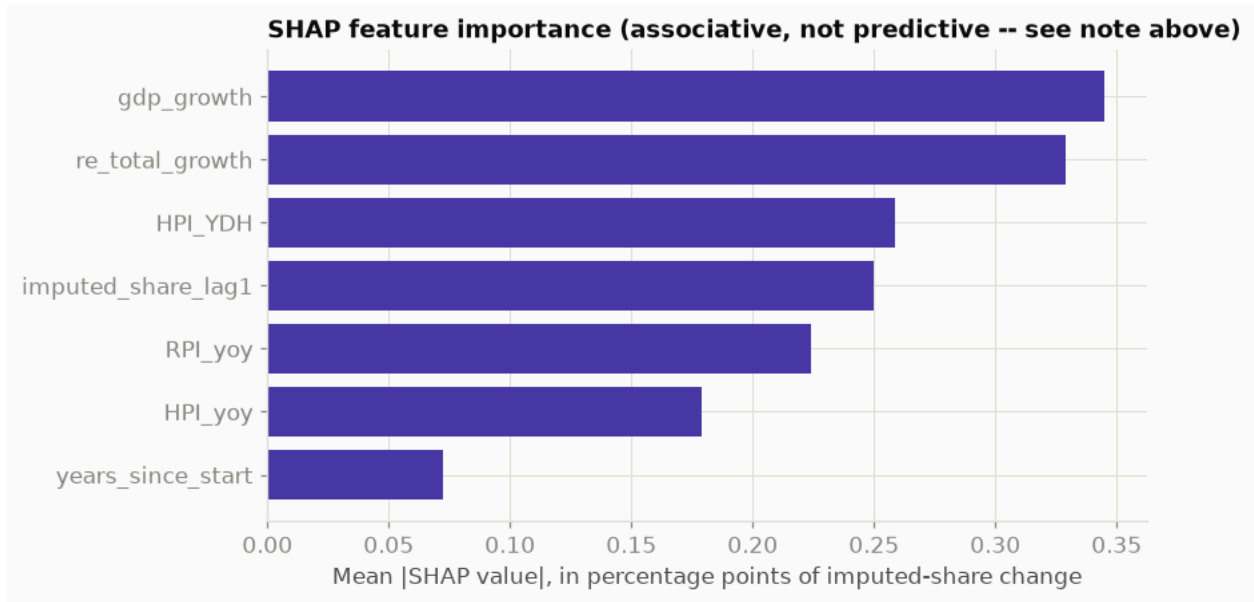
This means both relationships are real and statistically confirmed, but loose — a correlation of 0.30 or 0.39 means each factor explains well under a fifth of the variation in the other. Imputed rent growth does track real rent inflation *to some degree* — a reassuring basic sanity check — but the majority of its year-to-year movement comes from something else entirely (stock and quality adjustments, methodology updates, rebasing — see Section 3's Slovakia example). Crucially, how "pretend-rent-heavy" a country's books are tells you nothing about whether housing there is actually expensive relative to incomes — there is no meaningful relationship at all between the two.

We checked whether the "pretend rent" number moves in step with real rents that real tenants actually pay. It does, a little — like two neighbors who often, but not always, leave for work around the same time. Roughly 1 out of every 10 times the pretend-rent number jumps, that jump lines up with real rents moving too; the other 9 times, something else is behind it. And knowing how "pretend-rent-heavy" a country's real-estate books are tells you nothing about whether housing is actually expensive there relative to what people earn — a high pretend-rent share is not a red flag for affordability. It's just an accounting style.

9. Result 7 — Can Next Year's Move Be Predicted?

We tried to predict the year-on-year change in a country's imputed share (not its level, which barely moves and would make any model look artificially good) using contemporaneous house prices, rent prices, and GDP growth.

Figure 8. Which factors matter most, according to the model?



We tested three approaches — a linear regression, a gradient-boosted machine-learning model (XGBoost), and a compact neural network — against the simplest possible baseline: **just guess "no change from last year."**

Model	Cross-validated prediction error (percentage points)
Naive ("predict zero change")	1.41
Linear regression	1.47
XGBoost (gradient boosting)	1.48
Neural network	1.56

None of the more sophisticated approaches beat the simple "assume no change" guess. House-price growth, rent-price growth, and overall GDP growth do not reliably explain year-to-year movements in the imputed share. Combined with the Slovakia case study (Section 3), the most plausible explanation is that large moves in this number come from one-off statistical events — benchmark revisions, methodology changes, a country starting to report a new data split — rather than from smooth housing-market cycles that a model could learn to anticipate.

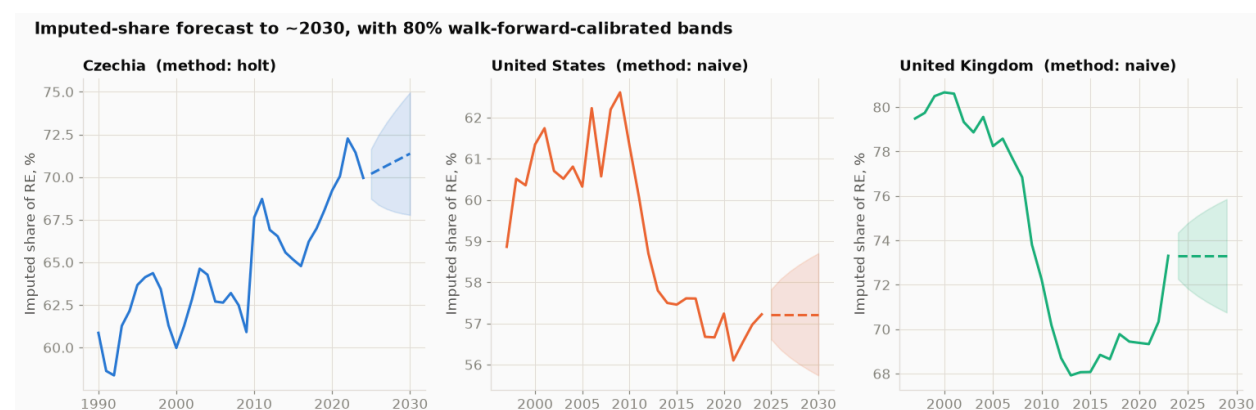
We fed a computer everything a market-savvy observer might reasonably guess would predict next year's pretend-rent number — house prices, rents, the overall economy — and asked it to bet on which way the

number would move next. The computer did no better than someone flipping a coin and guessing "no change." That's not the computer failing — it's a real finding: the size of next year's swing is not something the everyday housing market tells you in advance. It's far more likely to move because of a filing-cabinet decision than because of anything happening on the street.

10. Result 8 — An Honest Forecast to 2030

Given the finding above — that year-to-year changes behave close to random — we deliberately forecast using simple methods (last value, straight-line trend, and damped-trend smoothing), choosing whichever method has actually performed best for each country in a historical backtest, rather than picking the most sophisticated-looking option.

Figure 9. Imputed-share forecast to ~2030, with honestly-calibrated uncertainty bands



Country	2029/2030 forecast	80% likely range
Czechia	71.4%	67.8% – 75.0%
United States	57.2%	55.7% – 58.7%
United Kingdom	73.3%	70.8% – 75.9%

For the US and UK, the honest best guess is "roughly the same as today, give or take a few points" — because Section 9 showed we have no real ability to call the direction with confidence. For Czechia, the shaded uncertainty band widens noticeably by 2030, reflecting genuine uncertainty rather than false precision.

The dashed line and shaded cloud in each chart are our best honest guess for 2029/2030 — not a promise. For the US and UK we're essentially saying "probably about the same as today, give or take a few points." For Czechia, the shaded cloud widens a lot toward 2030 — picture a weather forecaster confidently naming tomorrow's temperature but only shrugging at "what will the weather be in six years." That widening band is intellectual honesty, not a flaw.

11. What This Means For You

If you rent your home

When you hear "the real estate sector grew X% this year," that number is *not* telling you landlords collected more rent, or that more rental homes got built — remember the ~65% median finding in Section 7. Look for news that separately reports rental-market or construction activity specifically, rather than the blended sector total. A country's imputed-rent share tells you nothing about whether your rent is becoming less affordable (Section 8 found no link to the price-to-income ratio) — don't treat "real estate GDP is booming" as a sign that your own rent is about to jump; watch the local Rent Price Index and vacancy rates instead, which are the direct measures.

If you own your home

Yes, statisticians count a "rent" you pay yourself — this does not put money in your pocket or take it out. It has zero effect on your mortgage, your taxes (in most countries), or your monthly budget. It only affects how big the national economy is reported to be. If your country's imputed-rent share is rising (Figure 6), that usually reflects your home's *assessed, notional value* rising faster than the going market rate for actual property services — not a signal about your personal home equity, which is driven by local sale prices, not this accounting split.

If you work in property (agents, landlords, developers, property managers)

Your slice of the numbers is specifically the "excluding imputed" line, not the total. When official statistics say "the property sector added \$X," your real business activity is best approximated by the *non-imputed* portion and by house-price/transaction indices (Section 8 shows these correlate with the non-imputed component better than with the total, $r \approx 0.39$). A rising total "real estate GDP" figure is not proof that business is picking up for you — Czechia's case (Section 6) shows the genuine services side can *shrink* even while the headline total grows. Track your own order book and transaction volumes, not the aggregate.

If you're a citizen following the news or voting on housing policy

Before accepting a politician's claim that "the property boom is driving the economy," ask the question this report is built around: *how much of that is pretend rent?* On the typical country's numbers, assume roughly two-thirds is, until shown otherwise. A sudden jump in a country's real-estate figures is more often a bookkeeping change (Section 3) than a sign that policy is working or the market has shifted — don't let a single year's jump drive your judgment of a government's housing record.

For policymakers and statistical communicators

Never cite "real estate contributed X% to GDP" without separating out imputed rent. Publish (or require statistical offices to publish) the imputed/non-imputed split as standard, the way some European countries already do. Before attributing a real-estate GDP jump to policy or market conditions, check whether a benchmark revision or methodology change occurred that year.

For journalists and economic analysts

Treat "real estate sector growth" headlines as a two-part number: report the non-imputed growth rate separately — it is the part that corresponds to jobs, transactions, and market activity. Use the *real (volume) decomposition* (Section 6), not the nominal share, to judge whether a country's real-estate sector is genuinely growing or merely being revalued.

For investors and market participants

A rising aggregate real-estate GDP figure is *not* a reliable signal of brokerage, construction, or transaction-market strength — check the non-imputed line specifically, or proxy it with transaction volumes / house-price indices.

12. Conclusions

1. **The structural finding is the headline, not the trend.** Across 43 countries, imputed rent makes up a **median of about 65%** of "real estate activities" value added. Whether that share is rising or falling locally is a second-order question next to the fact that **most of what gets reported as "real estate sector output" is a non-market valuation, everywhere, permanently** — not a recent drift.
2. **The trend is genuinely country-specific, not a global phenomenon.** Only 8 of 33 long-series countries show a statistically significant rising imputed share; a similar number show a significant *fall*; and the cross-country early-vs-late statistical test finds no significant average movement ($p \approx 0.23$; $p \approx 0.057$). Blanket claims that "housing GDP is increasingly fake everywhere" are not supported by the data — Czechia and the Netherlands fit that story; the UK and most of Scandinavia do not.
3. **But the *marginal* growth story leans one-sided, though not uniformly.** In every flagship country, the imputed component has contributed more than half of nominal real-estate-sector growth **since 2019** — 58.5% in the USA, 73.8% in Czechia, and 85.5% in the UK. The post-pandemic "real estate boom" reported in national accounts is, in Czechia and the UK, overwhelmingly a valuation effect in the most recent cycle; in the USA the imputed share of growth is closer to a bare majority.

4. **Real (volume) decomposition tells the sharpest story.** In Czechia, non-imputed real estate services have been *shrinking* in real terms ($-1.19\%/yr$) while imputed rent grows in real terms ($+2.03\%/yr$) — nominal sector growth sits on top of a contracting genuine service sector. The UK shows the reverse. This single result separates "growing" from "getting more expensive to value" more precisely than any share statistic can.
5. **Large moves are usually statistical events, not market events.** Slovakia's imputed share jumped from 0% to about 80% in one year purely because its statistical office began separately estimating it — a pure classification artifact indistinguishable, in the aggregate series, from a genuine boom.
6. **Year-to-year changes in the imputed share are not predictable from housing-market cyclicalities.** Three independent modeling approaches (linear regression, gradient boosting, and a neural network) all failed to beat a naive "no change" baseline out of sample. This is consistent with finding #5: big moves come from methodology events, not smooth market dynamics a model could learn.

13. Limitations

- The nominal-vs-real distinction relies on "previous-year prices" as an approximation, not a full official chain-linked volume series from national statistical offices.
 - Country coverage of the imputed/non-imputed split is uneven: 43 of roughly 60 countries in the underlying dataset report it at all, and among those, individual history lengths range from 2 to 35 years.
 - The house-price and rent-price indicators used for cross-checking are national-level, not segment-specific to the real-estate market — a proxy relationship, not a direct one.
 - The machine-learning and deep-learning results are genuinely negative, not merely modest: no usable predictive model for the imputed-share change was found, and this report says so plainly rather than reporting a weak model as if it were a meaningful finding.
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14. Suggestions for Further Research

- Add homeownership rate and construction-sector output as additional explanatory factors — these were not available in the dataset used here but are strong candidates for explaining the wide variation in imputed share seen across countries (Section 7), which housing-market cyclicalities alone does not explain.
- Extend the real (volume) decomposition in Section 6 to the full 43-country panel; it currently covers only the three flagship countries with sufficiently long price-adjusted series.

Appendix: Data Notes

- **Reliable panel:** 519 country-year rows across 43 countries, defined as rows where both the imputed-rent and non-imputed components were reported as strictly positive in the same year.
- **Countries with a partial reporting break** identified during data cleaning: Ireland and the Slovak Republic, both of which show a mix of zero-reported and non-zero years consistent with a statistical methodology change rather than a real economic shift (see Section 3).
- All charts and figures in this report were generated directly from the underlying OECD data; no figures were manually adjusted.